



Chapter 1

Matrix Algebra

MAE101 - ALGEBRA

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Matrix Algebra

1 Definition

An $n \times m$ matrix is a set of $n \times m$ real numbers arranged in n rows and m columns. A matrix A of order $n \times m$ is represented as follows

$$A_{n,m} = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,m} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,m} \end{pmatrix} = (a_{i,j}) \in \mathbb{R}^{n \times m}$$

where $a_{i,j}$ is the entry in the i^{th} row and the j^{th} column.

Zero matrix: is a matrix in which all elements are zero, $0_{n \times m}$ (or 0).

$$0_{3 \times 4} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Square matrix (ma trận vuông)

1 Definition

If the number of rows and columns is the same, namely $n = m$, then the matrix is called a **square matrix**.

$$A_{n,n} = A_n = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{pmatrix}$$

$M_n(\mathbb{R})$: the set of all n square matrices on $\mathbb{R}$.

The elements $a_{1,1}; a_{2,2}; \dots; a_{n,n}$ are called the **main diagonal** of A.

Example.

$$A_3 = \begin{pmatrix} -1 & 3 & 2 \\ 2 & -1 & 1 \\ 5 & 2 & 3 \end{pmatrix}$$

Special Square Matrix

1 Definition

1. An **upper triangular matrix U** is a square matrix in which all entries below the main diagonal are zero.
2. A **lower triangular matrix L** is a square matrix in which all entries above the main diagonal are zero.
3. A **diagonal matrix D** is a square matrix with zero entries except possibly on the main diagonal.
4. The **identity matrix I_n** of size n is the $n \times n$ square matrix with ones on the main diagonal and zeros elsewhere.

$$U = \begin{pmatrix} -1 & 3 & 2 \\ 0 & -1 & 1 \\ 0 & 0 & 3 \end{pmatrix}$$

$$L = \begin{pmatrix} 7 & 0 & 0 \\ 2 & -1 & 0 \\ 5 & 2 & -3 \end{pmatrix}$$

$$D = \begin{pmatrix} 9 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \text{diag}(9, -1, 3)$$

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$



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Equal Matrix

2 Matrix Operations

Two matrices are said to be equal if and only if they satisfy the following conditions:

- Both the matrices should have the same number of rows and columns.
- Both the matrices should have the same corresponding elements.

Example. Find x, y, z to

$$\begin{bmatrix} x+1 & 1 \\ 2x-1 & z \end{bmatrix} = \begin{bmatrix} 3y-4 & 1 \\ y-1 & 2z+2 \end{bmatrix}$$

Transpose matrix (ma trận chuyển vị)

2 Matrix Operations

For $A \in M_{n \times m}(\mathbb{R})$, we define the **transpose matrix** $A^T \in M_{m \times n}(\mathbb{R})$ of A , as the matrix whose row i is equal to column i of A .

Example.

$$A = \begin{bmatrix} 1 & -1 & 4 & 5 \\ 6 & -8 & 0 & 1 \\ 0 & 4 & -3 & 6 \end{bmatrix} \implies A^T = \begin{bmatrix} 1 & 6 & 0 \\ -1 & -8 & 4 \\ 4 & 0 & -3 \\ 5 & 1 & 6 \end{bmatrix}$$

Proposition

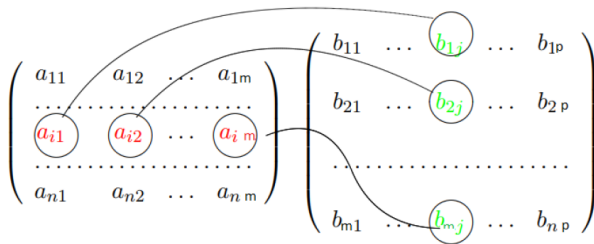
- If $A^T = A$, then A is a **symmetric matrix** (ma trận đối xứng).
- If $A^T = -A$, then A is a **skew symmetric matrix** (ma trận phản xạ).

Matrix Operations

2 Matrix Operations

1. **Scalar multiple:** If $A = [a_{i,j}]$, that is $kA = [ka_{i,j}]$
2. **Sum:** If $A = [a_{i,j}]$ and $B = [b_{i,j}]$, this take the form $A + B = [a_{i,j} + b_{i,j}]$.
3. **Product matrix:** Let $A \in M_{n \times m}(\mathbb{R})$ and $B \in M_{m \times p}(\mathbb{R})$. $C = A \cdot B$ is the matrix with n rows and p columns such that the element

$$c_{i,j} = a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \cdots a_{i,m}b_{m,j}$$



Matrix Operations - Example

2 Matrix Operations

Example 1. Given $A = \begin{bmatrix} 3 & 4 & 1 \\ 0 & 1 & -3 \end{bmatrix}$. Find matrix $2A$.

Example 2. Given $A = \begin{bmatrix} 2 & 1 & 3 \\ -1 & 2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 6 \end{bmatrix}$. Find $A + B$, $A - B$

Example 3. Given $A = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{bmatrix}$. Find $A \cdot B$, $B \cdot A$.

Matrix Operations - Example

2 Matrix Operations

Example 1. Given $A = \begin{bmatrix} 3 & 4 & 1 \\ 0 & 1 & -3 \end{bmatrix}$. Find matrix $2A$.

Example 2. Given $A = \begin{bmatrix} 2 & 1 & 3 \\ -1 & 2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 6 \end{bmatrix}$. Find $A + B$, $A - B$

Example 3. Given $A = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{bmatrix}$. Find $A \cdot B$, $B \cdot A$.

Solution.

$$AB = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix} \begin{bmatrix} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{bmatrix} = \begin{bmatrix} 2 \cdot 8 + 3 \cdot 7 + 5 \cdot 6 & 2 \cdot 9 + 3 \cdot 2 + 5 \cdot 1 \\ 1 \cdot 8 + 4 \cdot 7 + 7 \cdot 6 & 1 \cdot 9 + 4 \cdot 2 + 7 \cdot 1 \\ 0 \cdot 8 + 1 \cdot 7 + 8 \cdot 6 & 0 \cdot 9 + 1 \cdot 2 + 8 \cdot 1 \end{bmatrix} = \begin{bmatrix} 67 & 29 \\ 78 & 24 \\ 55 & 10 \end{bmatrix}$$

$\nexists BA$.

Sum - Theorem

2 Matrix Operations

Theorem

Let A, B and C denote arbitrary $n \times m$ matrices where n and m are fixed. Let k and p denote arbitrary real numbers. Then

1. $A + B = B + A$
2. $A + (B + C) = (A + B) + C$
3. There is an $n \times m$ matrix 0 , such that $0 + A = A$ for each A .
4. For each A there is an $n \times m$ matrix, $-A$, such that $A + (-A) = 0$
5. $k(A + B) = kA + kB$.
6. $(k + p)A = kA + pA$.
7. $(kp)A = k(pA)$.
8. $1A = A$.



Product matrix - Theorem

2 Matrix Operations

Theorem

Assume that a is any scalar, and that A, B , and C are matrices of sizes such that the indicated matrix products are defined. Then:

1. $IA = A$ and $AI = A$ where I denotes an identity matrix.
2. $A(BC) = (AB)C$.
3. $A(B + C) = AB + AC$
4. $(B + C)A = BA + CA$
5. $a(AB) = (aA)B = A(aB)$.
6. $(AB)^T = B^T A^T$.

k^{th} power of A

2 Matrix Operations

For a square matrix A and positive integer k , the k th power of A is defined by multiplying this matrix by itself repeatedly, that is

$$A^k = A \times A \times \cdots \times A$$

where there are k copies of the matrix A .

Example. For $A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$. Calculate A^2 and A^3 .

k^{th} power of A

2 Matrix Operations

For a square matrix A and positive integer k , the k th power of A is defined by multiplying this matrix by itself repeatedly, that is

$$A^k = A \times A \times \cdots \times A$$

where there are k copies of the matrix A .

Example. For $A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$. Calculate A^2 and A^3 .

Solution.

$$A^2 = AA = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix}$$
$$A^3 = A^2A = \begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 9 \\ 0 & 1 \end{bmatrix}$$



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Row-echelon form

3 Row-echelon form (dạng bậc thang)

A matrix is said to be in **row-echelon form** if it satisfies two conditions:

1. All zero rows (if exist) are at the bottom.
2. Each leading nonzero entry of a row is in a column to the right of the leading nonzero entry of the row above it.

Example.

- A row-echelon matrix $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}$, $\begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 1 \end{pmatrix}$.
- Not a row-echelon matrix $\begin{pmatrix} 0 & 0 & 0 \\ 3 & 1 & 4 \\ 0 & 0 & 5 \end{pmatrix}$, $\begin{pmatrix} 0 & 2 & 7 \\ 0 & 3 & 4 \\ 0 & 0 & 5 \end{pmatrix}$, $\begin{pmatrix} 1 & 3 & 5 \\ 0 & 0 & 4 \\ 2 & 1 & 3 \end{pmatrix}$.

Reduced row-echelon form

3 Row-echelon form (dạng bậc thang)

If a matrix in echelon form satisfies the following additional conditions, then it is in reduced row-echelon form (dạng bậc thang rút gọn):

1. The leading entry in each nonzero row is 1.
2. Each leading 1 is the only nonzero entry in its column.

Example. A row-echelon matrix

$$\begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



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Elementary row operations

4 Elementary row operations

The three elementary row operations are:

1. (Row Swap) Exchange any two rows: $r_i \leftrightarrow r_j$
2. (Scalar Multiplication) Multiply any row by a constant: $r_i := \alpha r_i$, $\alpha \neq 0$.
3. (Row Sum) Add a multiple of one row to another row: $r_i := r_i + \beta r_j$, $\beta \neq 0$.

Example. Use elementary row operations to transform A to B.

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & -2 & 3 \\ 3 & -1 & 2 \end{pmatrix}, B = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 1 & -7/5 \\ 0 & 0 & 0 \end{pmatrix}.$$

Rank of the matrix

4 Elementary row operations

The number of non-zero rows in the row-echelon form of a matrix A is called the rank of the matrix, denoted as $\text{rank}(A) = r$.

Example. Determine the rank of matrix

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & -3 \\ 3 & -1 & 0 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 9 \\ 3 & 6 & 9 & 11 \\ 4 & 8 & 14 & 10 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 3 \\ 5 & 12 & 2 & 7 \end{pmatrix}$$

$$D = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix}$$



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Invertible matrix (ma trận khả nghịch)

5 Matrix Inverses

If A is a square matrix, a matrix B is called an inverse of A if and only if

$$AB = BA = I_n$$

A matrix A that has an inverse is called an invertible matrix.

Find An Inverse Matrix by Using Elementary row operations

$$(A \mid I_n) \xrightarrow{\varphi_1} (A_1 \mid B_1) \rightarrow \dots \xrightarrow{\varphi_p} (I_n \mid A^{-1})$$

Example 1. Show that $B = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$ is an inverse of $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$.

Example 2. Find A^{-1} (if it exists) with $A = \begin{pmatrix} 1 & -3 \\ 2 & 2 \\ \frac{2}{3} & -1 \end{pmatrix}$ and $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}$

Matrix Inverses - Theorem

5 Matrix Inverses

Theorem

All the following matrices are square matrices of the same size.

1. I is invertible and $I^{-1} = I$.
2. If A is invertible, so A^{-1} , and $(A^{-1})^{-1} = A$.
3. If A and B are invertible, so is AB , and $(AB)^{-1} = B^{-1}A^{-1}$.
4. If $A_1, A_2, \dots, A_k$ are all invertible, so is their product $A_1A_2 \dots A_k$, and

$$(A_1A_2 \dots A_k)^{-1} = A_k^{-1} \dots A_2^{-1}A_1^{-1}$$

5. If A is invertible, so is A^k for any $k \geq 1$, and $(A^k)^{-1} = (A^{-1})^k$.
6. If A is invertible and $a \neq 0$ is a number, then aA is invertible and $(aA)^{-1} = \frac{1}{a}A^{-1}$.
7. If A is invertible, so is its transpose A^T , and $(A^T)^{-1} = (A^{-1})^T$.



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Matrix equations

6 Matrix equations

Let $A, A' \in M_n(\mathbb{R})$ be invertible and $B \in M_{n \times p}(\mathbb{R})$, $C \in M_{m \times n}(\mathbb{R})$, $D \in M_n(\mathbb{R})$.
Then

1. $AX = B \iff X = A^{-1}B$
2. $XA = C \iff X = CA^{-1}$
3. $AXA' = D \iff X = A^{-1}DA'^{-1}$

Example. Solve the equation

a. $\begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix} X = \begin{bmatrix} -2 & 3 \\ 2 & 5 \end{bmatrix}$

b. $X \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 3 \\ 2 & 5 \end{bmatrix}$

c. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix} X \begin{bmatrix} 3 & 2 \\ 3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 3 & 1 \\ 2 & -1 \end{bmatrix}$



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Linear transformation

7 Transformation

Theorem

If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation, then for each $k = 1, 2, \dots$

$$T(a_1x_1 + a_2x_2 + \dots + a_kx_k) = a_1T(x_1) + a_2T(x_2) + \dots + a_kT(x_k)$$

for all scalars a_i and all vectors x_i in $\mathbb{R}^n$.

Example. $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation, $T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ and

$$T \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}, \text{ find } T \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

Linear transformation

7 Transformation

Theorem

If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation, then for each $k = 1, 2, \dots$

$$T(a_1x_1 + a_2x_2 + \dots + a_kx_k) = a_1T(x_1) + a_2T(x_2) + \dots + a_kT(x_k)$$

for all scalars a_i and all vectors x_i in $\mathbb{R}^n$.

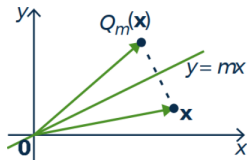
Example. $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation, $T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ and

$$T \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}, \text{ find } T \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

Solution. $T \begin{bmatrix} 4 \\ 3 \end{bmatrix} = T \left[\frac{11}{3} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 1 \\ -2 \end{bmatrix} \right] = \frac{11}{3} \begin{bmatrix} 2 \\ -3 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 5 \\ 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 27 \\ -32 \end{bmatrix}$

Reflection

7 Transformation



The reflection $Q_m : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ in the line $y = mx$ has matrix

$$\frac{1}{1+m^2} \begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}$$

Example.

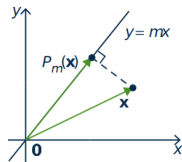
We denote by $Q_m : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ reflection in the line $y = x$. We see that

$$Q_m \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} b \\ a \end{bmatrix}$$

Q_m is a matrix transformation

Projection

7 Transformation



The projection $P_m : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ on the line $y = mx$ has matrix

$$\frac{1}{1+m^2} \begin{bmatrix} 1 & m \\ m & m^2 \end{bmatrix}$$

Example.

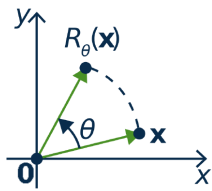
Let $P_m : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be projection on the line $x - 2y = 0$. Find $P_m[-5, 6]^T$.

Solution. We have

$$P_m \begin{bmatrix} -5 \\ 6 \end{bmatrix} = \begin{bmatrix} 0.8 & 0.4 \\ 0.4 & 0.2 \end{bmatrix} \begin{bmatrix} -5 \\ 6 \end{bmatrix} = \begin{bmatrix} -1.6 \\ -0.8 \end{bmatrix}$$

Rotation of $\mathbb{R}^2$

7 Transformation



The rotation $R_\theta : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ counterclockwise about the origin through the angle θ has matrix

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Example.

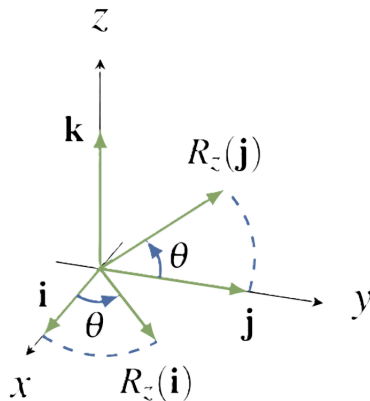
We denote by $R_\pi : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ counterclockwise rotation about the origin through an angle of π . We see that

$$R_\pi \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -a \\ -b \end{bmatrix}$$

R_π is a matrix transformation

Rotation of $\mathbb{R}^3$

7 Transformation



Let $R_{z,\theta} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ denote rotation of $\mathbb{R}^3$ about the z - *axis* through an angle θ from the positive x - *axis* toward the positive y - *axis* has matrix

$$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$



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Problems

8 Problems

1. Let $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 1 \\ 3 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 3 & -4 \\ -1 & 2 & 1 \end{bmatrix}$.

Compute the matrix $2A - B^T$, AB , BA , AC , CC^T , C^TC , A^3 , B^2A^T .

2. Suppose that A and B are $n \times n$ matrices. Simplify the expression

a. $(A + B)^2 - (A - B)^2$ b. $A(BC - CD) + A(C - B)D - AB(C - D)$

3. Let $A = \begin{bmatrix} 3 & 1 & 2 \\ 4 & 8 & 0 \\ 0 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 5 & 2 & 1 \\ 1 & 8 & 0 & -6 \\ 1 & 4 & 3 & 7 \end{bmatrix}$. Compute AB .

4. Find the inverse of each of the following matrices.

a. $\begin{bmatrix} 1 & 5 \\ 2 & -1 \end{bmatrix}$ b. $\begin{bmatrix} 1 & -1 & 2 \\ -5 & 7 & -11 \\ -2 & 3 & -5 \end{bmatrix}$

Problems

8 Problems

5. Given $A^{-1} = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 5 \\ -1 & 1 & 0 \end{bmatrix}$. Find a matrix X such that

a. $AX = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$

b. $AX = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

c. $XA = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 1 \end{bmatrix}$

6. Find A when

a. $(3A)^{-1} = \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}$

b. $(A^{-1} - 2I)^T = -2 \begin{bmatrix} 1 & 4 \\ 3 & 11 \end{bmatrix}$

7. Find A^{-1} if

a. $A^2 - 6A + 5I = 0$

b. $A^2 + 3A - I = 0$

c. $A^4 = I$



Problems

8 Problems

8. Solve for X (where A, B and C are $n \times n$ invertible matrices)

a. $\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} X = \begin{bmatrix} 1 & -1 \\ 3 & 3 \end{bmatrix}$ b. $ABXC = B^T$ c. $AX^TBC = B$

9. Compute $\begin{bmatrix} -1 & 3 \\ 0 & 1 \end{bmatrix}^{101}$

10. Find the $(2, 1)$ -entry of the product $\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 2 & 5 & 1 \\ 4 & -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 & 2 & 1 \\ 2 & 3 & 2 \\ 5 & 1 & 0 \\ 0 & 4 & 3 \end{bmatrix}$

11. Compute the rank of each of the following matrices.

a. $\begin{bmatrix} 1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$ b. $\begin{bmatrix} 1 & 1 & -1 & 3 \\ -1 & 4 & 5 & -2 \\ 1 & 6 & 3 & 4 \end{bmatrix}$

Problems

8 Problems

12. Determine the values of m such that the rank of the matrix is 2.

a. $\begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & 5 \\ 1 & 2 & m \end{bmatrix}$

b. $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 1 & 1 & 5 \\ -3 & 6 & 1 & m \end{bmatrix}$

13. Find all values of m for which the matrix is invertible $\begin{bmatrix} m & 1 & 3 \\ 1 & 3 & 2 \\ 1 & 4 & 5 \end{bmatrix}$.

14. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation, and assume that $T(1, 2) = (-1, 1)$ and $T(0, 3) = (-3, 3)$.

a. Compute $T(11, -5)$

b. Compute $T(1, 11)$

c. Find the matrix of T

15. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation such that the matrix of T is $\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$.

Find $T(3, 2)$.



Q&A

Thank you for listening!